Summary of the Project

This project focuses on understanding whether unstructured data such as news, themes, and earnings can add meaningful insight into market behaviour. Instead of building a complex model, the approach is centred on analysing different signals extracted from text and evaluating their relationship with market movement. The analysis follows a structured process where each signal is tested individually and then in combination to understand its effectiveness, with the objective of identifying which signals provide useful understanding of market behaviour.

Key Findings

The main objective of this project was to understand whether unstructured data such as news, themes, and earnings can provide meaningful insight into market behaviour beyond traditional structured data.

From the overall analysis, one clear pattern emerged. Not all signals extracted from unstructured data are equally useful. Sentiment, which captures only the tone of the news, turned out to be a weak signal, showing around a 50% alignment with market movement. This indicates that tone alone is not sufficient to explain market behaviour.

Entity-based analysis improved the understanding slightly by focusing on the subjects of the news. However, most entities did not show a consistent or stable relationship with market returns, which suggests that individual elements of the news often act as noise when analysed in isolation.

The most meaningful results came from theme-based analysis. Themes capture broader narratives by combining multiple related signals into one structure. These showed clearer and more interpretable patterns in both positive and negative directions, indicating that market behaviour is better understood at a narrative level rather than through isolated signals.

When earnings data was analysed, it showed a strong positive bias, as companies tend to present their performance favourably. On its own, earnings data did not provide strong alignment with market behaviour. However, when combined with theme signals, there was a noticeable improvement at the stock level, with alignment reaching around 70%.

At the market level, however, all signals remained close to a 50% outcome, showing that broader market movement is not easily explained using these signals alone

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What Worked

The most effective part of the project was the shift from isolated signals to broader thematic understanding.

Theme-based analysis worked relatively well because it captures how multiple factors interact together, which is closer to how market behaviour actually evolves. It reduced the noise seen in sentiment and entity analysis and provided more interpretable insights.

Combining theme signals with earnings data also worked to some extent at the company level. When company communication aligned with broader market themes, the signal became more meaningful, and this was reflected in better alignment with stock movement.

Another important aspect that worked well was the structured approach of breaking the problem into smaller questions and analysing each signal independently before combining them. This helped in understanding where the actual strength and weakness of each signal lies.

What Did Not Work

Sentiment and entity signals did not perform well as standalone indicators. Sentiment lacked depth, and entity analysis, even after filtering, remained inconsistent and noisy.

The multi-signal approach, where sentiment, entity, and theme were combined together, did not improve the results. Instead, it introduced more noise and reduced the effectiveness of the stronger theme signal. This shows that combining multiple weak signals does not necessarily create a stronger one.

Earnings data also had limitations due to its positive bias and limited availability. Many earnings events occurred on non-trading days, reducing the usable data further.

At the market level, none of the signals — individual or combined — were able to move beyond a mid-level outcome. This indicates that market behaviour depends on many external factors that are not captured in the dataset.

Final Takeaway

The project shows that unstructured data can add meaningful context to market behaviour, but it is not sufficient to explain or predict market movement in a consistent way.

Themes emerge as the most useful signal because they capture broader narratives, while earnings data can add value at the company level when aligned with these themes.

However, the effectiveness of these signals depends heavily on data quality, availability, and how the signals are constructed. Limited data, bias in text, and subjective methodology reduce the strength of the analysis.

In conclusion, unstructured data should not be seen as a replacement for structured data, but as a complementary layer that provides context. It helps in understanding *why* the market behaves in a certain way, but not reliably *what* the market will do next.
